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CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 03 – General Proficiency

1 hour

04 JUNE 2007 (a.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **In addition to the 1 hour, candidates are allowed a reading time of 10 minutes. Writing may begin during the 10-minute period.**

2. **This paper is divided into THREE sections. Candidates MUST answer ONE question from EACH section. Where appropriate, answers should be illustrated by diagrams.**

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SECTION A

Answer ONE question from this section.

1. (a) With the aid of a diagram, describe how the rib cage **and** lung function during the breathing process. **(8 marks)**

(b) (i) Well-trained athletes recover more quickly after completing a race than untrained athletes. Give TWO reasons why this is so.

(ii) Athletes are usually advised against smoking cigarettes. Suggest TWO reasons for such advice. **(8 marks)**

(c) Although iron is a requirement for all mature females, it is recommended that female athletes consume more iron than females who are not athletes. Suggest TWO reasons for this recommendation. **(4 marks)**

Total 20 marks

2. (a) (i) With the aid of a diagram, describe how the process of meiosis occurs.

(ii) Give TWO important differences between meiosis and mitosis. **(8 marks)**

(b) Scientists establish relationships among individuals using their knowledge of genetics. Construct a sequence of genetic diagrams to illustrate how scientists can use knowledge of genetics to show how a couple with blood types A and B respectively can produce a baby with blood type O. **(8 marks)**

(c) Sugarcane farmers usually grow a single variety of the plant and spend a lot of time, money and effort on protection from insect infestation.

(i) Suggest TWO reasons why sugarcane farmers tend to choose a single variety for their crops.

(ii) What are the implications to the sugarcane farmers of insect infestation of their crop? **(4 marks)**

Total 20 marks

SECTION B

Answer ONE question from this section.

3. (a) (i) Using ONLY a fully annotated diagram of the internal structure of a leaf, explain the role of the parts involved in photosynthesis. Include in EACH annotation the name of the structure and its function.

(ii) Write a balanced equation that effectively summarises photosynthesis.

(14 marks)

(b) (i) One effect of hurricanes is the complete removal of foliage (leaves) from trees. Many trees seem to readily survive the loss of all their leaves, even though they are the main photosynthesising organs. Explain how this is possible.

(ii) Gardeners sometimes remove leaves and branches from trees with the expectation of improving their yield. Suggest why this is a good practice.

(6 marks)

Total 20 marks

4. (a) (i) Using ONLY a fully annotated diagram show how the human digestive system is able to effectively perform its function.

(ii) Suggest ONE way in which the digestive system protects itself from disease-causing organisms in food. **(10 marks)**

(b) Identify TWO ways in which the body defends itself against disease organisms **and** explain the importance of the body having different ways of defending itself from disease. **(4 marks)**

(c) Most countries are concerned about an outbreak of a communicable disease, even though the disease, for example, 'bird flu', occurs in a distant country and in a species that is not human.

(i) What is a communicable disease?

(ii) Explain why countries should be concerned about <u>any</u> outbreak of a communicable disease. **(6 marks)**

Total 20 marks

SECTION C

Answer ONE question from this section.

5. (a) (i) What are decomposers?

(ii) Explain the role of decomposers in the carbon cycle.

(iii) Identify TWO types of decomposers **and** describe the characteristics they possess that make them suited to their role as decomposers. **(12 marks)**

(b) Excessive amounts of gases such as carbon dioxide in the atmosphere have resulted in increased trapping of heat energy around the earth, causing higher global temperatures (global warming) similar to that which occurs under greenhouse conditions.

(i) What are the benefits of growing plants under greenhouse conditions?

(ii) Suggest TWO reasons why global warming is NOT generally desirable.

(iii) Suggest ONE way in which global warming may be advantageous.

(8 marks)

Total 20 marks

6. (a) Using an example in EACH case, distinguish between the following pairs of terms used in the study of ecology.

- Biotic and physical
- Food chain and food web
- Community and population
- Habitat and environment

(8 marks)

(b) Wildlife reserves have become popular tourist destinations in several Caribbean territories. Wildlife authorities must protect plants and animals while providing opportunity for people to benefit from the natural beauty of the reserves.

Suggest THREE biological factors that wildlife authorities should consider before giving the public access to a wetland (swamp) reserve. For EACH factor you identify suggest what precautions the authorities may take in allowing public access.

(9 marks)

(c) In one Caribbean territory, visitors to the coral reef were allowed to experience its beauty by snorkeling, swimming and walking on the reef with plastic shoes. Within two decades, the reef deteriorated drastically and visitors were no longer allowed to walk on the reef.

What are the MAIN lessons to be learnt from this example? **(3 marks)**

Total 20 marks

END OF TEST